

Name of Student: Vyankatesh Suresh Rohokale	PRN: 26070122518
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 15	
TITLE : Exception Handling using Finally , Throw , and Throws	

Problem Statement

Write a Java program to demonstrate exception handling using finally , throw , and throws.

Learning Objectives

To implement finally , throw , and throws for effective exception management.

Questions :

1. Create a Login program that throws an exception for invalid password and uses

2. finally block.

Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

Programme

1:

```
J Demo.java
1  import java.util.*;
2
3  class Login {
4
5      void login(String password) {
6
7          if (!password.equals("java123")) {
8              throw new IllegalArgumentException("Invalid password");
9          }
10
11         System.out.println("Login successful");
12     }
13 }
```

```

14
15 public class Demo {
16
17     public static void main(String[] args) {
18
19         Scanner sc = new Scanner(System.in);
20
21         System.out.print("Enter password: ");
22         String password = sc.nextLine();
23
24         try {
25             Login l = new Login();
26             l.login(password);
27         }
28         catch (IllegalArgumentException e) {
29             System.out.println("Error: " + e.getMessage());
30         }
31         finally {
32             System.out.println("Login verification completed.");
33         }
34
35         sc.close();
36     }
37 }

```

2:

```

Demo.java
1 import java.util.*;
2 class ATM {
3     void verifyPIN(int pin) {
4
5         if (pin != 1234) {
6             throw new IllegalArgumentException("Invalid PIN");
7         }
8
9         System.out.println("PIN verified successfully");
10    }
11 }
12
13 public class Demo {
14
15     public static void main(String[] args) {
16
17         Scanner sc = new Scanner(System.in);
18
19         System.out.print("Enter PIN: ");
20         int pin = sc.nextInt();
21
22         try {
23             ATM atm = new ATM();

```

```

24         atm.verifyPIN(pin);
25     }
26     catch (IllegalArgumentException e) {
27         System.out.println("Error: " + e.getMessage());
28     }
29     finally {
30         System.out.println("PIN verification process completed.");
31     }
32
33     sc.close();
34 }
35 }

```

Output

1:

```

• piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter password: 123
Error: Invalid password
Login verification completed.
• piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter password: java123
Login successful
Login verification completed.
❖ piyushrohokale@Piyushs-MacBook-Air DSA assignments % █

```

2:

```

• piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter PIN: 123
Error: Invalid PIN
PIN verification process completed.
• piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter PIN: 1234
PIN verified successfully
PIN verification process completed.
❖ piyushrohokale@Piyushs-MacBook-Air DSA assignments % █

```

Github:

Java_assignments / assignment15 /

VyankateshRohokale

Add ATM PIN verification functionality

388adc5 · 4 minutes ago History

Name	Last commit message	Last commit date
..		
.DS_Store	a14	10 minutes ago
1.png	a14	10 minutes ago
11.png	a14	10 minutes ago
2.png	a14	10 minutes ago
22.png	a14	10 minutes ago
ATMPINVerification.java	Add ATM PIN verification functionality	4 minutes ago
Login.java	Create Login.java	4 minutes ago

Conclusion:

The programs successfully demonstrate exception handling using finally and throw in Java. The Login and ATM PIN Verification programs explicitly throw exceptions for invalid inputs and handle them using catch. The finally block executes after the verification process, ensuring that the completion message is displayed regardless of whether an exception occurs.